

Lab 09: WLAN vs Wireless Concepts

Comparing Wired and Wireless Network Configurations

Lab #9

Topic: Wireless
Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build and compare a wired LAN topology side-by-side with a wireless WLAN topology.
- Identify the differences in hardware, configuration, and behaviour between wired and wireless.
- Configure both topologies to reach the same server and compare results.
- Understand 802.11 standards (a/b/g/n/ac) and their differences.
- Relate to Network+ objective: 2.4 – Explain wireless standards and technologies.

2. Topology Comparison

Feature	Wired LAN	Wireless WLAN
Medium	Cat5e/Cat6 Ethernet cable	Radio frequency (2.4GHz / 5GHz)
Standard	IEEE 802.3	IEEE 802.11 a/b/g/n/ac
Speed	Up to 1 Gbps (1000BASE-T)	Up to 1.3 Gbps (802.11ac)
Connectivity	Switch port + RJ-45	Access Point + wireless NIC
Security	Physical cable access	WPA2/WPA3 encryption required
Mobility	Fixed (cable length)	Mobile within AP coverage area
Interference	Minimal	Susceptible to RF interference

3. Wired LAN Configuration

Left side of topology: PCs connected via switch to a server.

Device	Interface	IP Address	Subnet Mask	Gateway
Wired-PC0	Fa0	172.16.1.10	255.255.255.0	172.16.1.1
Wired-PC1	Fa0	172.16.1.11	255.255.255.0	172.16.1.1
Server0	Fa0	172.16.1.100	255.255.255.0	172.16.1.1

Router0	Fa0/0	172.16.1.1	255.255.255.0	—
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4. Wireless WLAN Configuration

Right side: laptops connecting wirelessly via an Access Point to the same network.

SSID	WLAN-Lab
Security	WPA2-PSK
Passphrase	Network+2024
AP IP	172.16.1.2
Laptop0 IP	172.16.1.20 (or DHCP)
Laptop1 IP	172.16.1.21 (or DHCP)

4.1 802.11 Standards Reference

Standard	Frequency	Max Speed	Range	Notes
802.11b	2.4 GHz	11 Mbps	35m	Legacy, prone to interference
802.11g	2.4 GHz	54 Mbps	38m	Backward compatible with b
802.11n	2.4/5 GHz	600 Mbps	70m	MIMO, dual-band
802.11ac	5 GHz	1.3 Gbps	35m	Wi-Fi 5, beamforming
802.11ax	2.4/5/6 GHz	9.6 Gbps	30m	Wi-Fi 6, OFDMA

5. Verification

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Wired-PC0> ping 172.16.1.100    ! Wired PC → Server
Laptop0>   ping 172.16.1.100    ! Wireless client → Server
Laptop0>   ping 172.16.1.10     ! Wireless → Wired (cross-medium)
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6. Key Takeaways

- WLAN (Wireless LAN) is a LAN implemented with wireless technology — it still connects to the same IP subnet.
- Wired connections are more reliable and faster; wireless provides mobility and flexibility.
- Both wired and wireless clients can coexist on the same subnet and communicate freely.
- Wi-Fi security is critical — never leave a network open; always use WPA2 or WPA3.
- The 5 GHz band is faster and less congested than 2.4 GHz but has shorter range.